

Strength of Materials



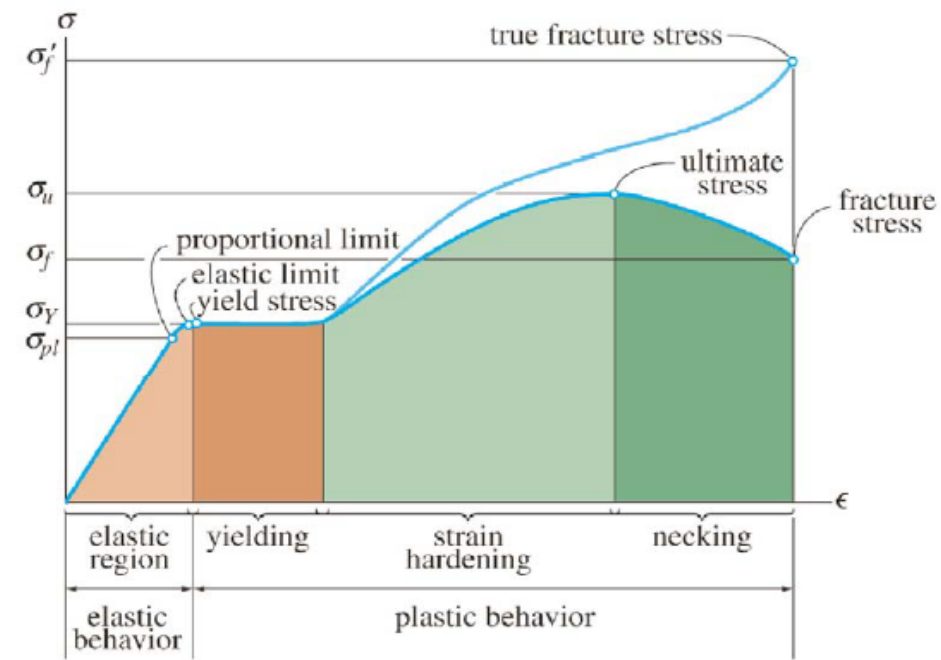
Chapter One: Stress & Strain

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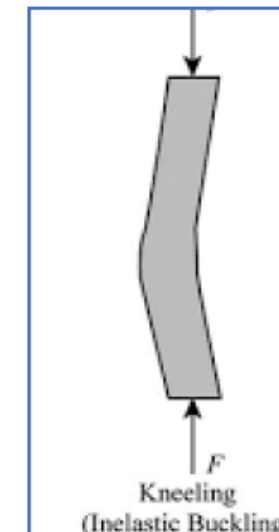
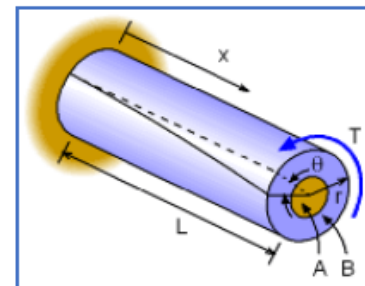
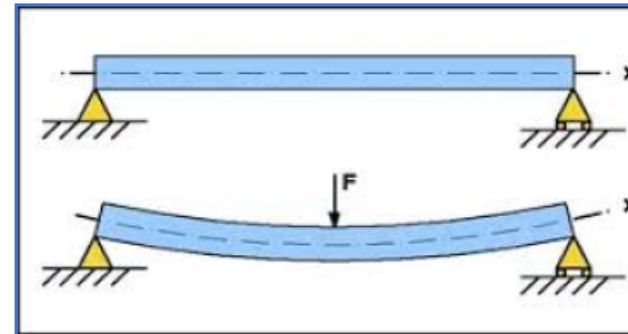
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Definition

- Mechanics of materials is a branch of applied mechanics that deals with the behavior of solid bodies subjected to various types of loading.
- Other names for this field of study are **strength of materials** and **mechanics of deformable bodies**.
- The solid bodies include bars with axial loads, **shafts** in torsion, **beams** in bending, and **columns** in compression.

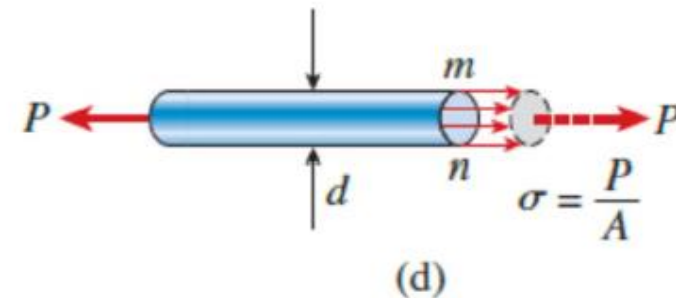
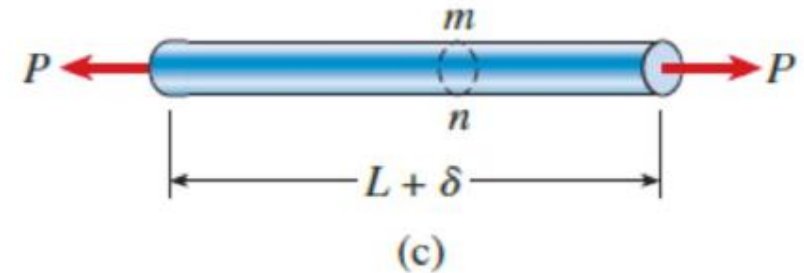
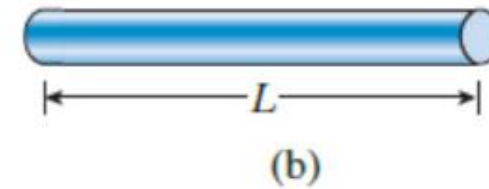
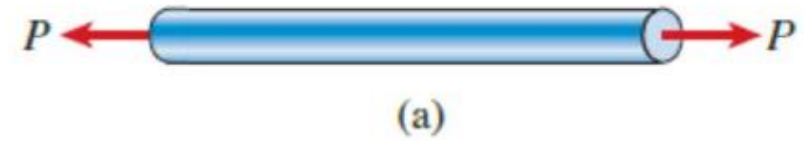


Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)



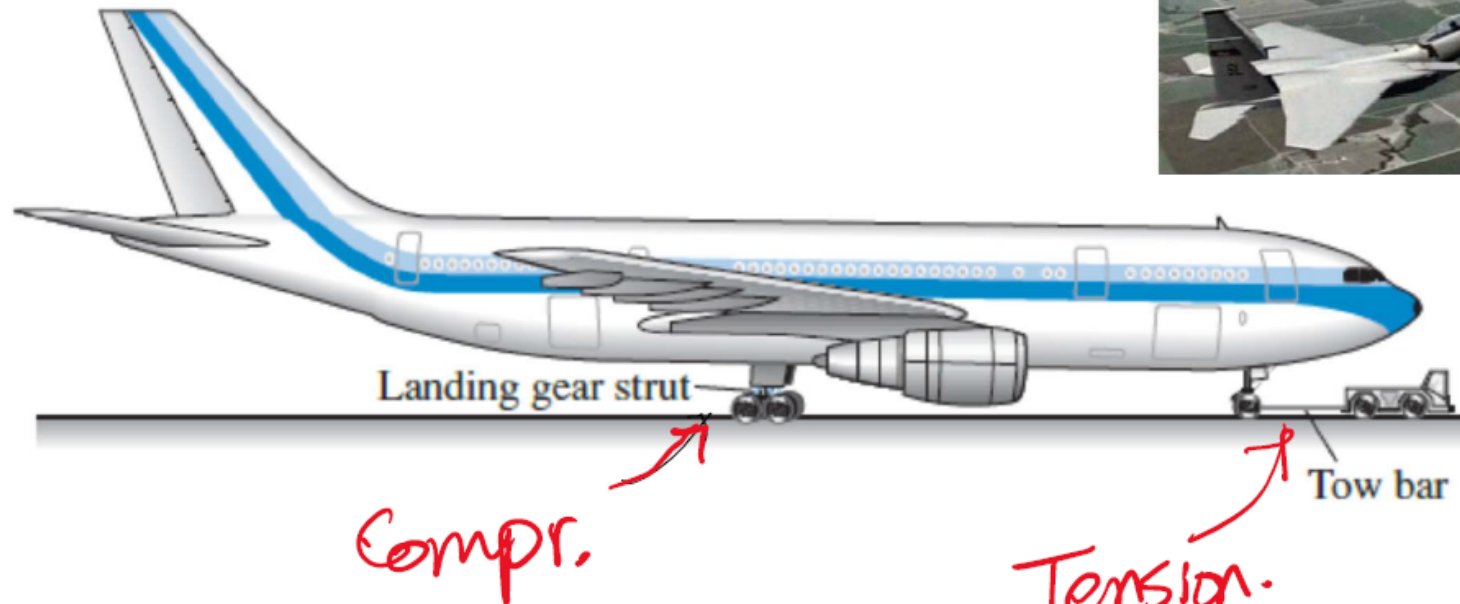
Objective of Mechanics of Materials

- The principal objective of mechanics of materials is to determine the stresses, strains, and displacements in structures and their components due to the loads acting on them.
- If we can find these quantities for all values of the loads up to the loads that cause failure, we will have a complete picture of the mechanical behavior of these structures.



The Need of Mechanics of Materials

- An understanding of mechanical behavior is essential for **the safe design of all types of structures**, whether airplanes and antennas, buildings and bridges, machines and motors, or ships and spacecraft.
- That is why mechanics of materials is a basic subject in so many engineering fields.



- Statics and dynamics are also essential, but those subjects deal primarily with the **forces** and **motions** associated with particles and rigid bodies.
- In mechanics of materials, we go one step further by examining the **stresses** and **strains** inside real bodies, that is, bodies of finite dimensions that deform under loads.
- To determine the stresses and strains, we use the **physical** properties of the materials as well as numerous **theoretical laws and concepts**.

Theoretical analyses and experimental results

- Theoretical analyses and experimental results have equally important roles in mechanics of materials.
- We use theories to derive **formulas** and **equations** for predicting mechanical behavior, but these expressions cannot be used in practical design unless the physical properties of the materials are known.
- Such properties are available only after careful experiments have been carried out in the laboratory.
- Not all practical problems are amenable to theoretical analysis alone, and in such cases physical testing is a necessity.

The Historical Development of Mechanics of Materials

The historical development of mechanics of materials is a fascinating blend of both **theory and experiment**—theory has pointed the way to useful results in some instances, and experiment has done so in others.

Leonardo da Vinci (1452–1519) and **Galileo Galilei** (1564–1642) performed experiments to determine the strength of **wires, bars, and beams**, although they did not develop adequate theories (by today's standards) to explain their test results.

Leonardo da Vinci
(1452–1519)



Galileo Galilei
(1564–1642)



- The famous mathematician **Leonhard Euler** (1707–1783) developed the mathematical theory of columns and calculated the critical load of a column in 1744, long before any experimental evidence existed to show the significance of his results.
- Without appropriate tests to back up his theories, Euler's results remained unused for over a hundred years, although today they are the basis for the design and analysis of most columns.



Review of Pre-Requisites

Mechanics of Materials

- Three Fundamental areas of engineering mechanics
 - Statics
 - Dynamics
 - Strength of Materials
- Statics and dynamics are devoted primarily to study of the **external effects** of forces on rigid bodies, that is , bodies for which the change in shape (deformation) can be neglected.
 - In Contrast
- Strength of materials deals with the relations between externally applied loads and their internal effects on bodies. Moreover, the bodies are no longer assumed to be rigid; deformations, however small, are of major interest.

- In **Mechanical Design**, the engineer must consider both dimensions and material properties to satisfy requirements of strength and rigidity. When loaded, a machine part or structure should neither break, nor deform excessively.
- The difference between **rigid-body** mechanics and **strength of materials** can be further emphasized by considering the following example.



- Study of the Bar
 - Statics
 - Strength of Materials

- What may happen if Engineering Mechanics isn't applied properly?



$$\Sigma F=0$$

$$\Sigma M=0$$

Establish
equilibrium

Equilibrium:

- In many applications of mechanics, the sum of the forces acting on a body is zero, and a state of equilibrium exists.

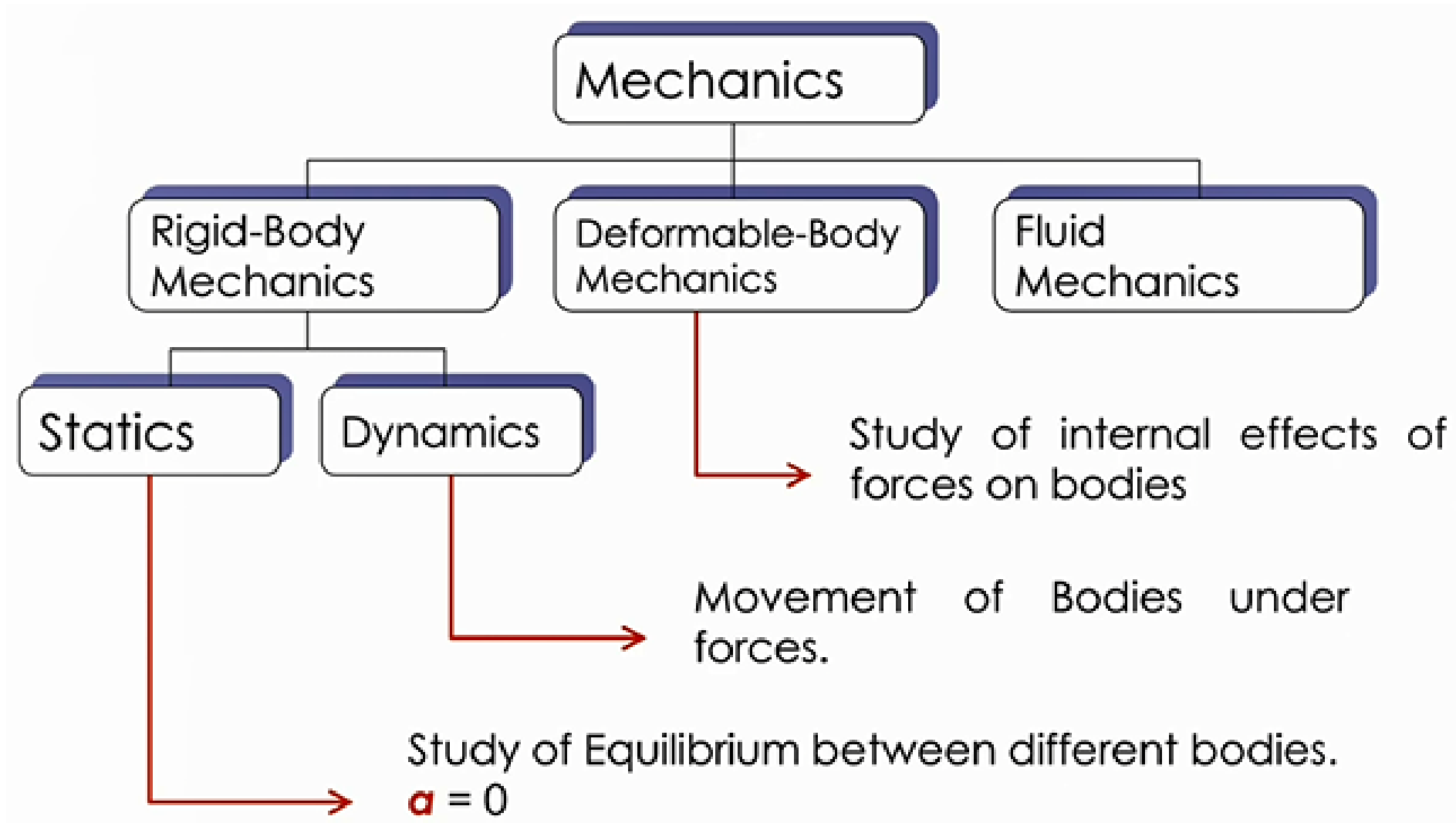
$$\Sigma F=0$$

$$\Sigma M=0$$

- A body is in equilibrium if all forces and moments applied to it are in balance.
- Or if the resultant of all forces and moments acting on the body is zero.



Mechanics

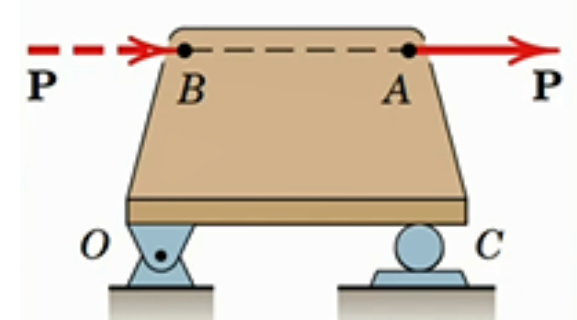


Force

- Force: The action of one body on another body. Its is a vector quantity.

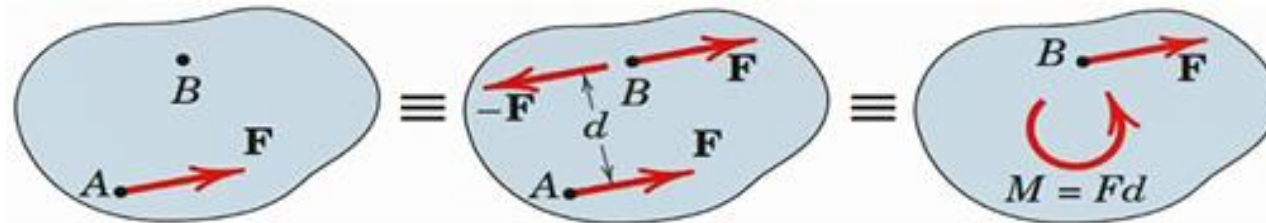
Principle of Transmissibility:

A force may be applied to any point on its given line of action without altering the resultant effects of the force *external* to the rigid body on which it acts



Force-Couple Systems

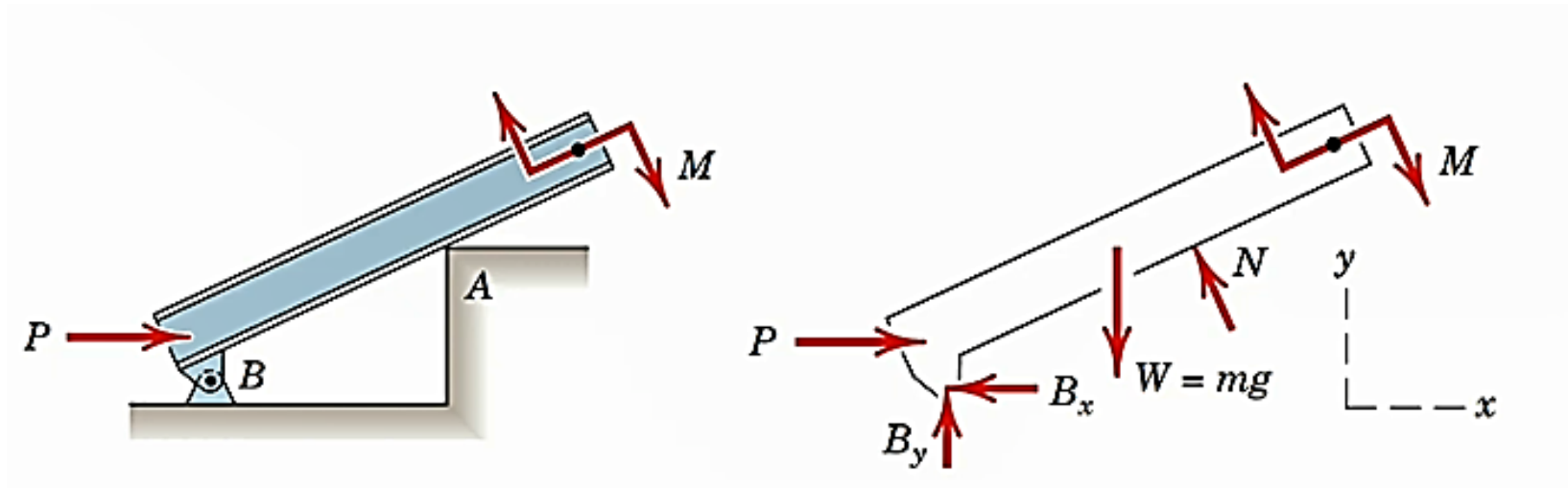
A force can be replaced by an equal parallel force and a couple in the same plane. This combination of a force and a couple is called *Force-Couple System*.



Free Body Diagram

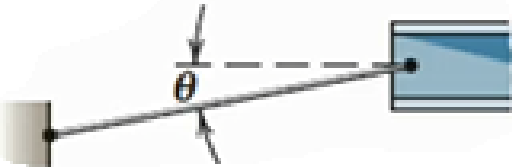
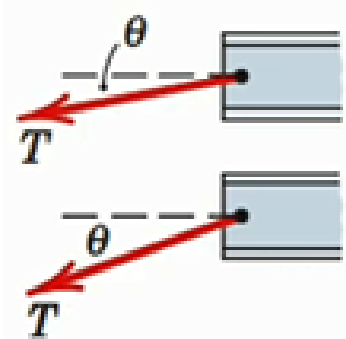
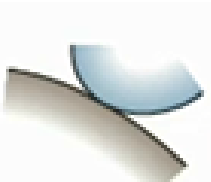
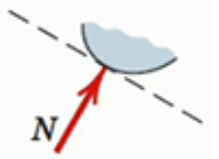
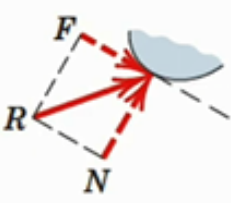
System Isolation and the Free Body Diagram

- A *mechanical system* is defined as a body or group of bodies which can be conceptually isolated from all other bodies
- A diagrammatic representation of an isolated system, forces and reaction forces on the system is called *Free Body Diagram*.



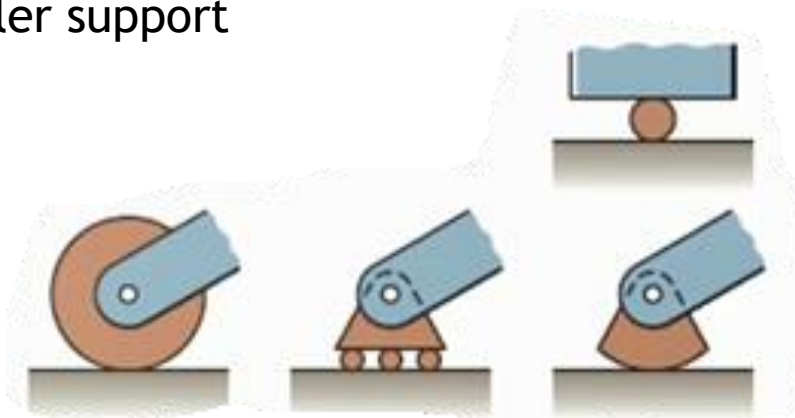
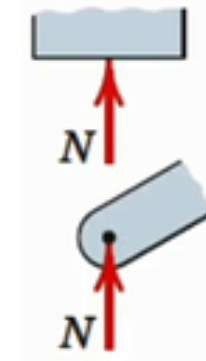
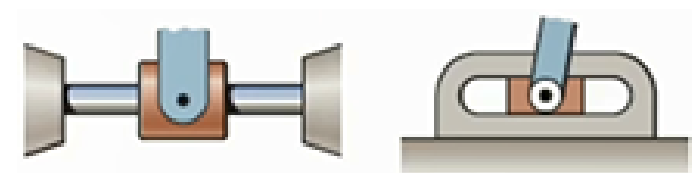
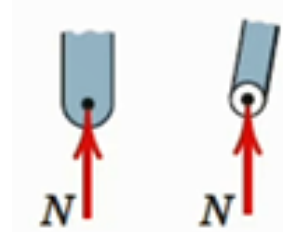
Free Body Diagram

Modeling the action of Forces

Type of contact and Force Origin	Action on Body to Be Isolated
1. Flexible cable, belt, chain, or rope Weight of cable negligible  Weight of cable not negligible 	Force exerted by a flexible cable is always a tension away from the body in the direction of the cable 
2. Smooth Surfaces 	Contact force is compressive and is normal to the surface. 
3. Rough Surfaces 	Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant 

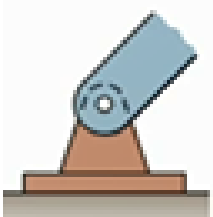
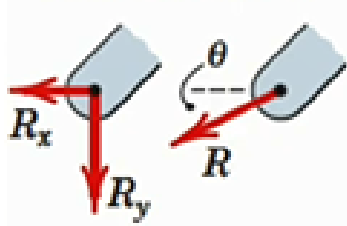
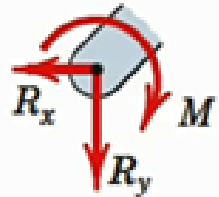
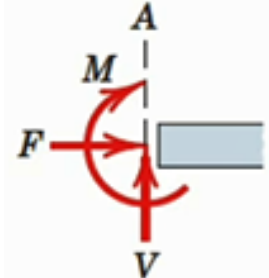
Free Body Diagram

Modeling the action of Forces

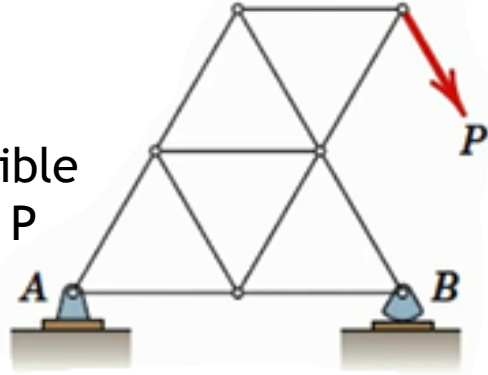
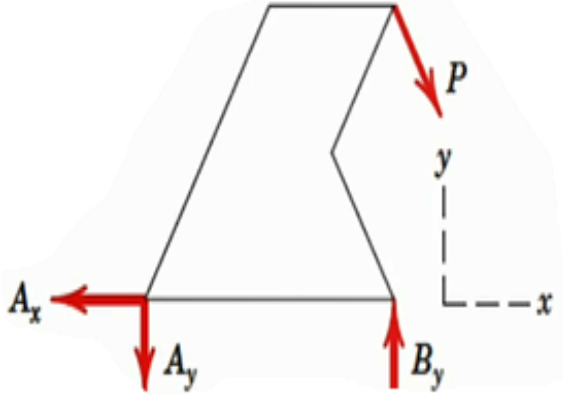
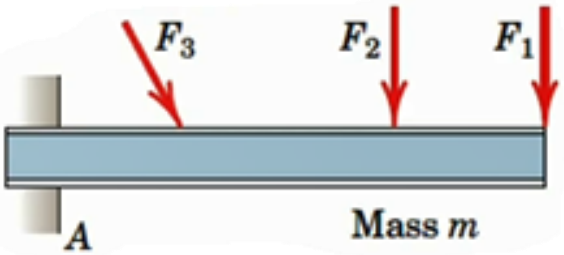
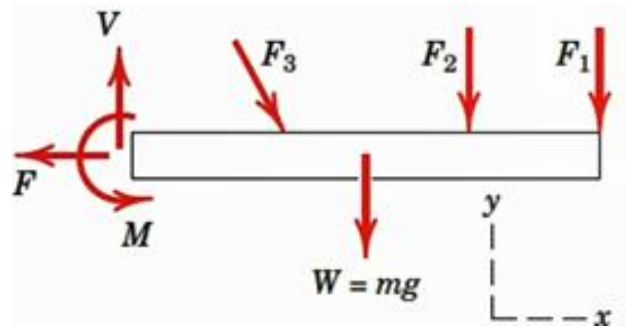
Type of contact and Force Origin	Action on Body to Be Isolated
4. Roller support 	Roller, rocker, or ball support transmits a compressive force normal to the supporting surface. 
5. Freely sliding guide 	Collar or slider free to move along smooth guides; can support force normal to guide only. 

Free Body Diagram

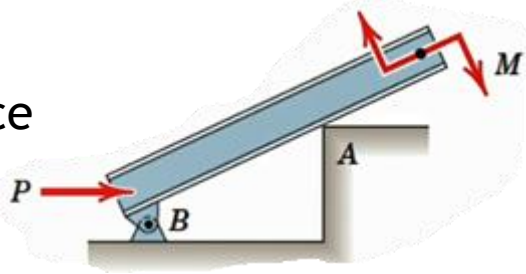
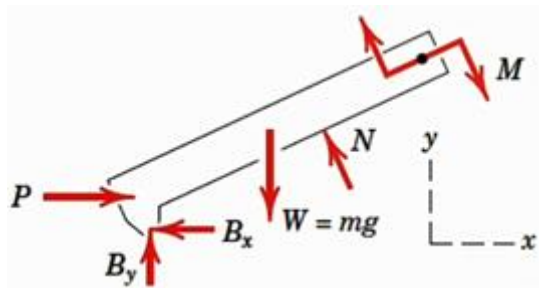
Modeling the action of Forces

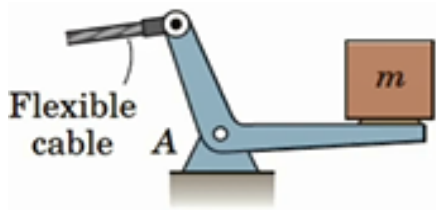
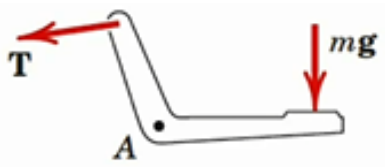
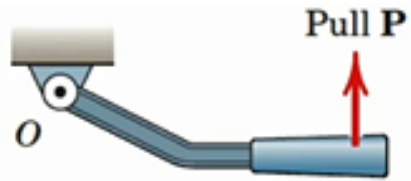
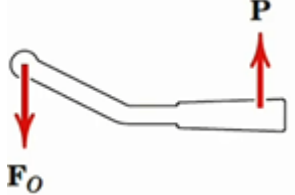
Type of contact and Force Origin	Action on Body to Be Isolated	
6. Pin connection 	Pin free to turn  Pin not free to turn 	A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis. We may either show two components R_x and R_y or a magnitude R and direction θ. A pin not free to turn also supports a couple M
7. Built-in or fixed support 		A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation.

Free Body Diagram (*Examples*)

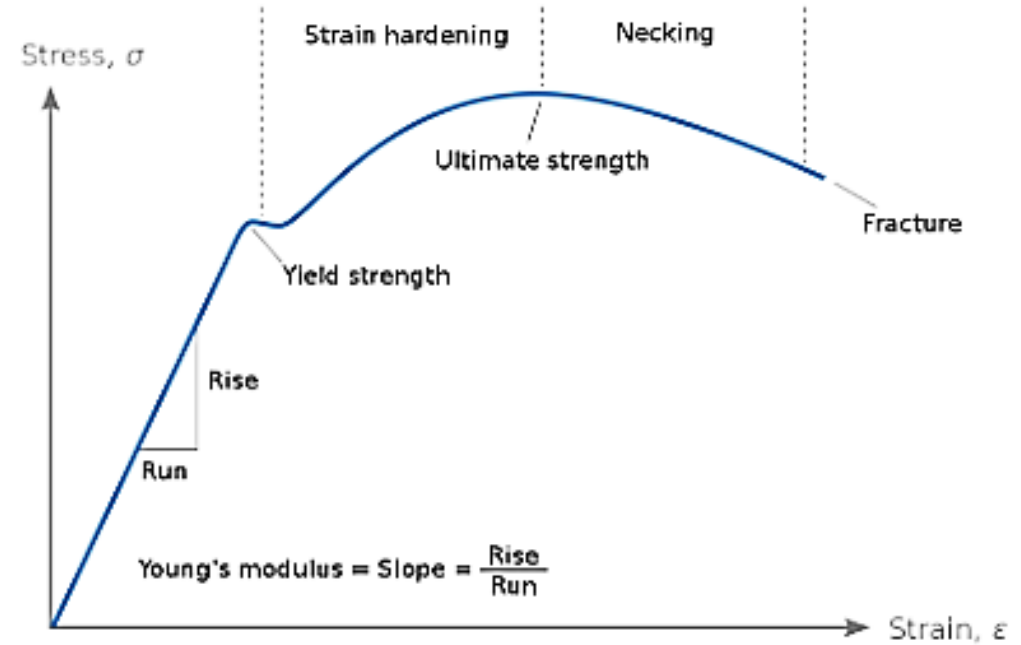
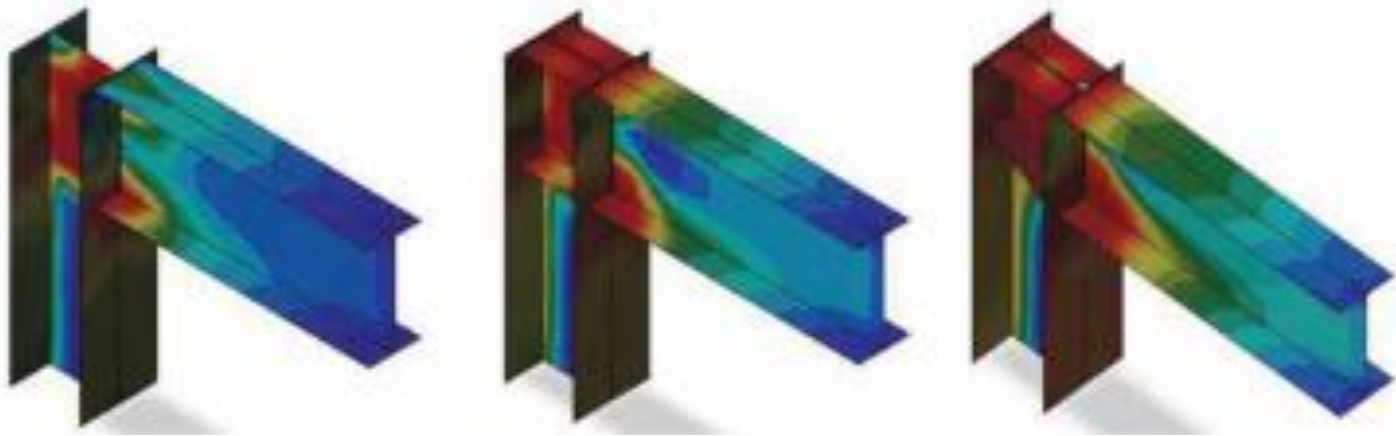
Sample Free-Body Diagrams	
Mechanical System	Free-Body Diagram of Isolated Body
1. Plane truss Weight of truss Assumed negligible Compared with P 	
2. Cantilever beam 	

Free Body Diagram (*Examples*)

Sample Free-Body Diagrams	
Mechanical System	Free-Body Diagram of Isolated Body
3. Beam Smooth surface contact at A. Mass m 	

	Body	Incomplete FBD
1. Bell crank Support mass m with pin support at A.		
2. Control lever Applying torque to shaft at O.		

Chapter One: Stress and Strain



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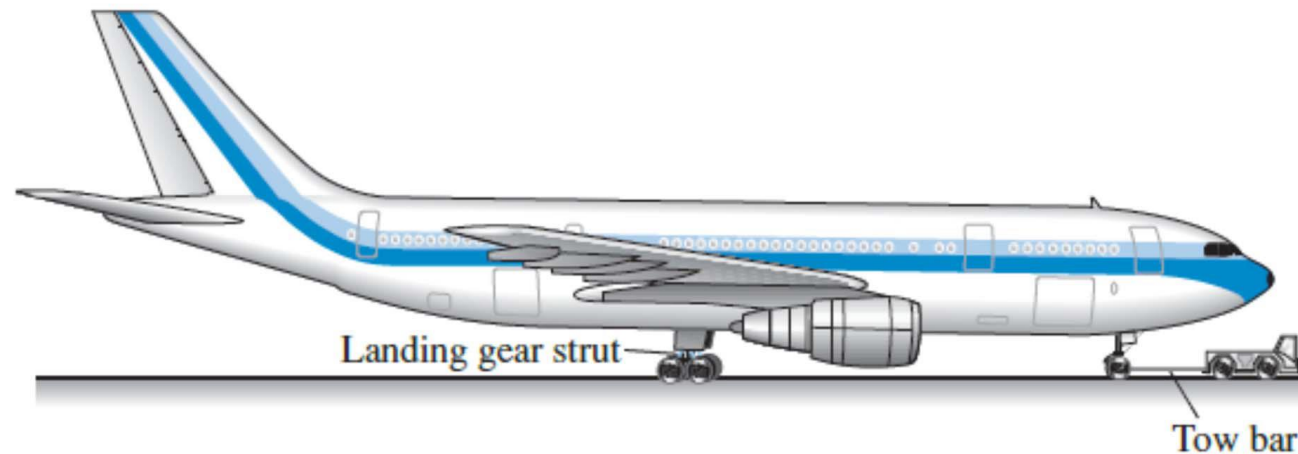
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- Pin Bearing Stresses
- Stress in Two Force Members
- Stress on an Oblique Plane
- Maximum Stresses
- Stress under General Loadings
- State of Stress
- Factor of Safety

Introduction

The most fundamental concepts in mechanics of materials are stress and strain. These concepts can be illustrated in their most elementary form by considering a prismatic bar subjected to axial forces.

A prismatic bar is a straight structural member having the same cross section throughout its length, and an axial force is a load directed along the axis of the member, resulting in either tension or compression in the bar.

Examples are shown



loads

The external forces acting on bodies are known as “*loads*”. These external forces can either be *tensile* or *compressive*



Tension force present in towing ropes or lifting hoists.



Compression occurs in legs of chair/bench as one sit on it.

Types of loads

Different No. of ways in which load(s) can be applied to a member:

1. **Static or Dead loads:** i.e. non fluctuating loads, those generally caused by gravity
2. **Live loads:** as trucks crossing a bridge.
3. **Impact or Shock loads:** caused by sudden blows.
4. **Fatigue, fluctuating or alternating loads:** i.e. the magnitude and sign of the load change with time.

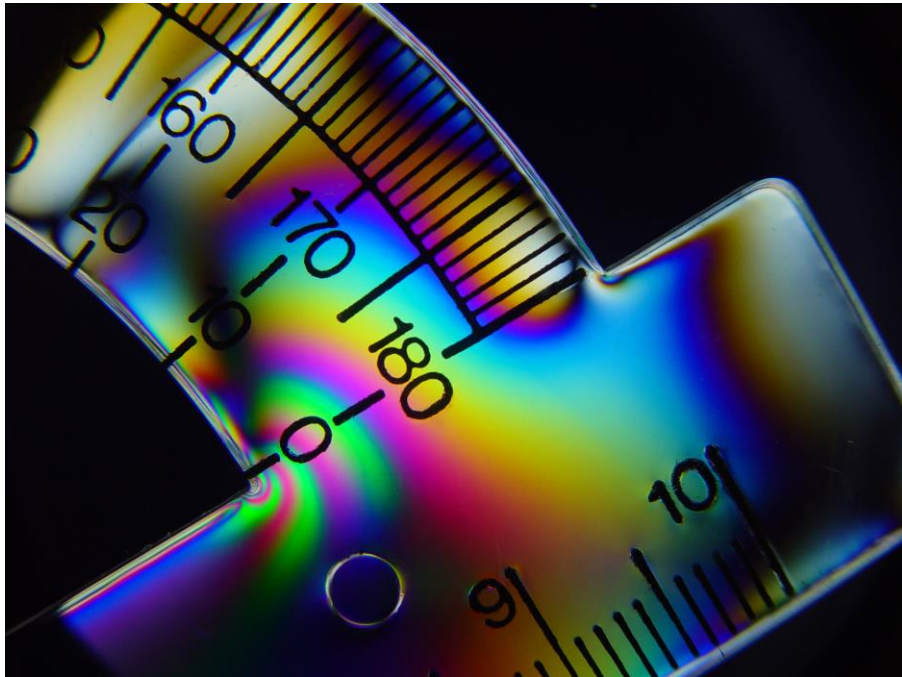
Loads may be due to any one of the reason as:

- Due to environment in which the component works.
- Through contact with other members.
- Due to fluid pressures.
- Due to gravity or inertia forces.

We can analyze a body only if it is under *static equilibrium*

Concept of Stress

As we know that in mechanics of **deformable solids**, externally applied forces act on a body and body suffers deformation. From equilibrium point of view, this action would be opposed or reacted by internal forces which are set up with the particles of material due to cohesion. These internal forces give to a **Concept of Stress**.



Intensity of **Internal resistive forces** is known as stress.

Or

Internal Resistance(per unit area) to the external forces is called **Stress**

Axial loads

Assumptions:

- Uniform and prismatic(straight) bar, rod, tube etc.
- Homogeneous & Isotropic material
- Load 'P' directed axially along the centroidal axis of cross section.



Outline - Stresses

Categories of stresses

- **Direct Stresses**
 - Normal
 - Shear
- **Torsional Stresses**
- **Bending Stresses**
- **Thermal Stresses**
- **Combined Stresses**

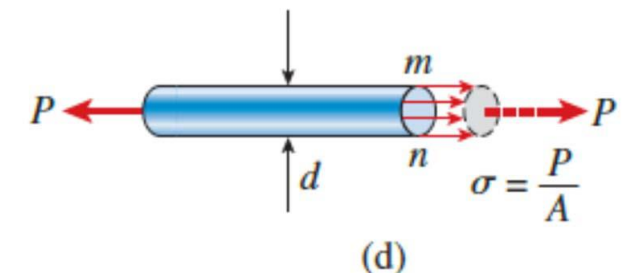
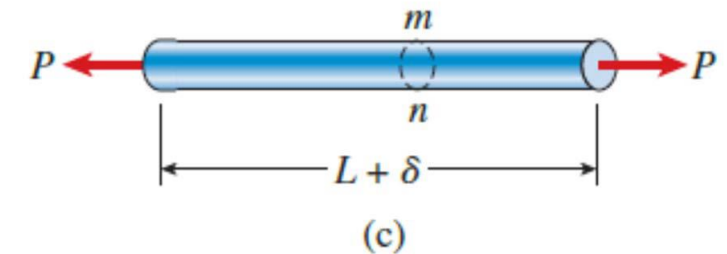
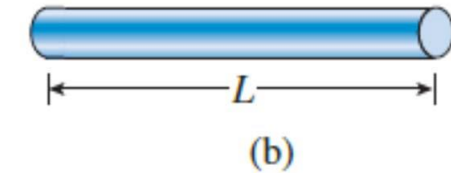
Prismatic bar in tension

We disregard the weight of the bar itself and assume that the only active forces are the axial forces P at the ends.

Prismatic bar in tension:

- (a) free-body diagram of a segment of the bar,
- (b) segment of the bar before loading,
- (c) segment of the bar after loading, and
- (d) normal stresses in the bar

- At the right-hand end of this free body (section mn) we show the action of the removed part of the bar (that is, the part to the right of section mn) upon the part that remains.
- This action consists of continuously distributed stresses acting over the entire cross section, and the axial force P acting at the cross section is the resultant of those stresses. (The resultant force is shown with a dashed line)



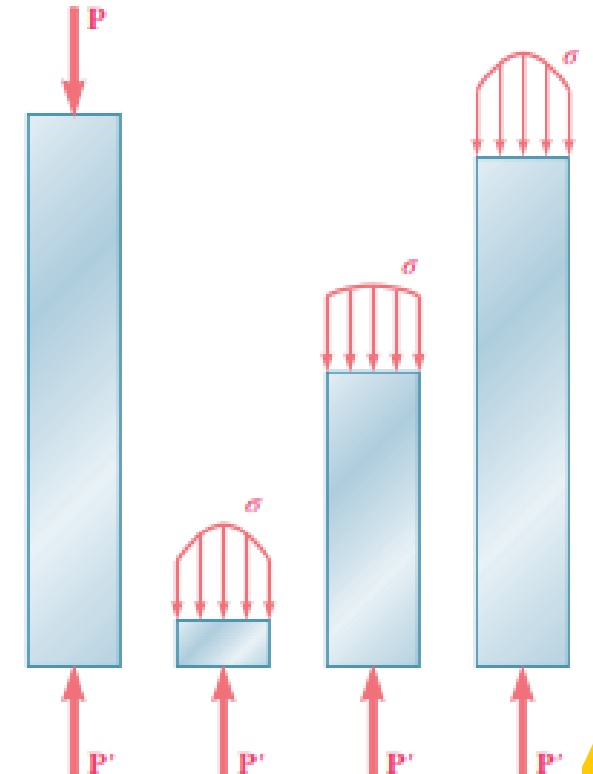
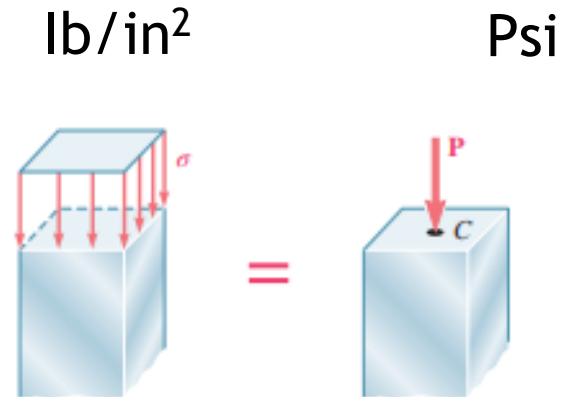
Normal Stresses

- If the load is perpendicular (normal) to the area concerned.

$$\sigma = \frac{P}{A}$$

- stress σ will be expressed in N/m² pascal (Pa).

- Kilopascal (kPa)
- Megapascal (Mpa)
- Gigapascal (GPa):



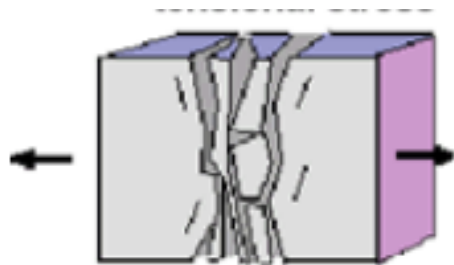
Tensile or Compressive Stresses

The normal stresses can be either **tensile** or **compressive** whether the stresses acting of the area or into the area.

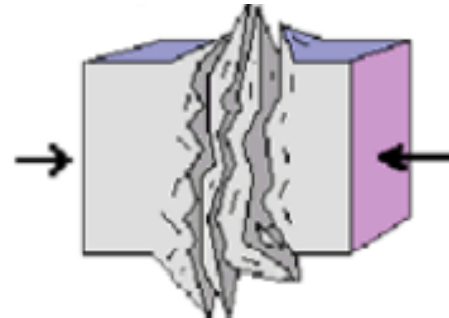
Sign Convention:

- Tensile: +ve
- Compressive: -ve

tensional stress

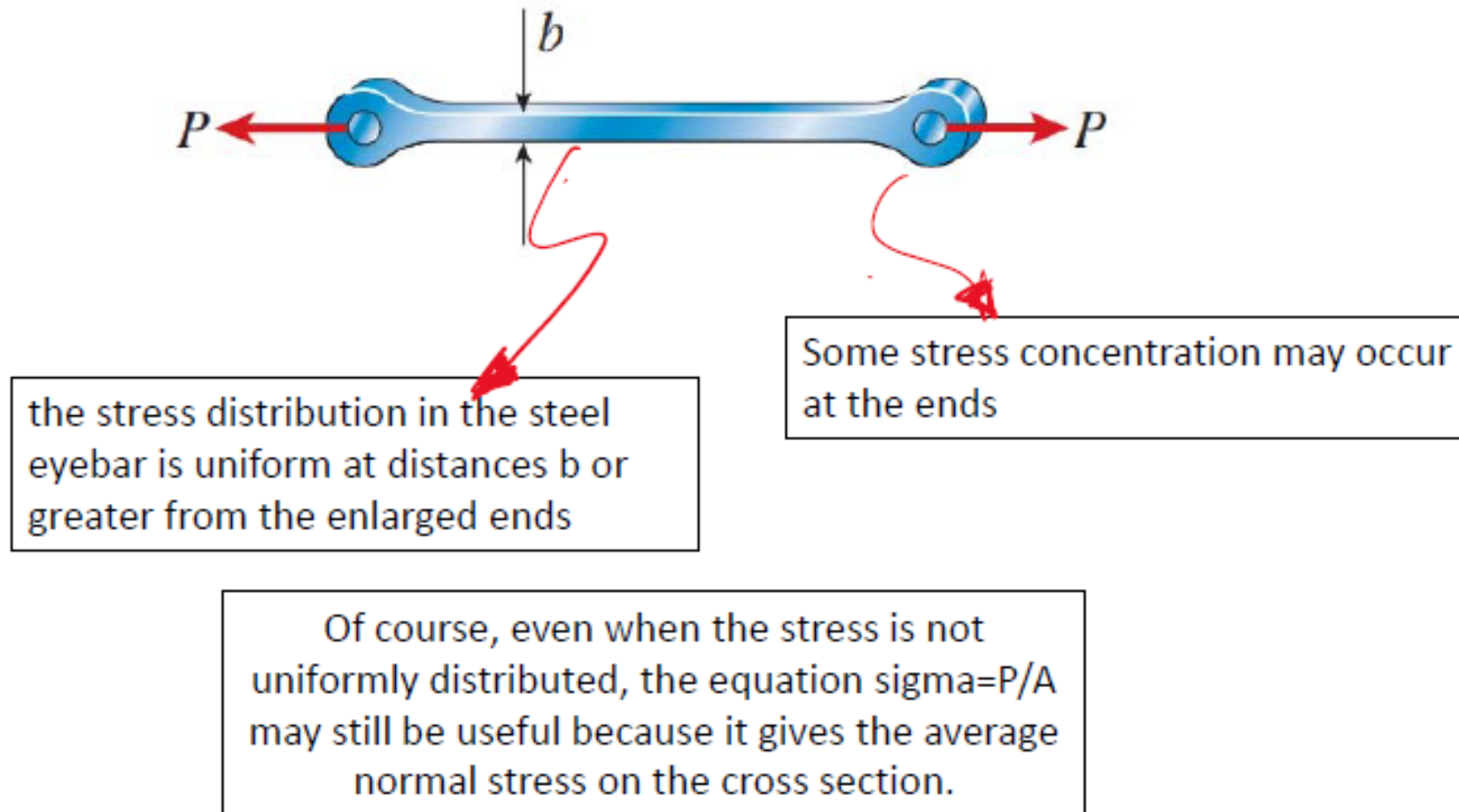


compressional stress

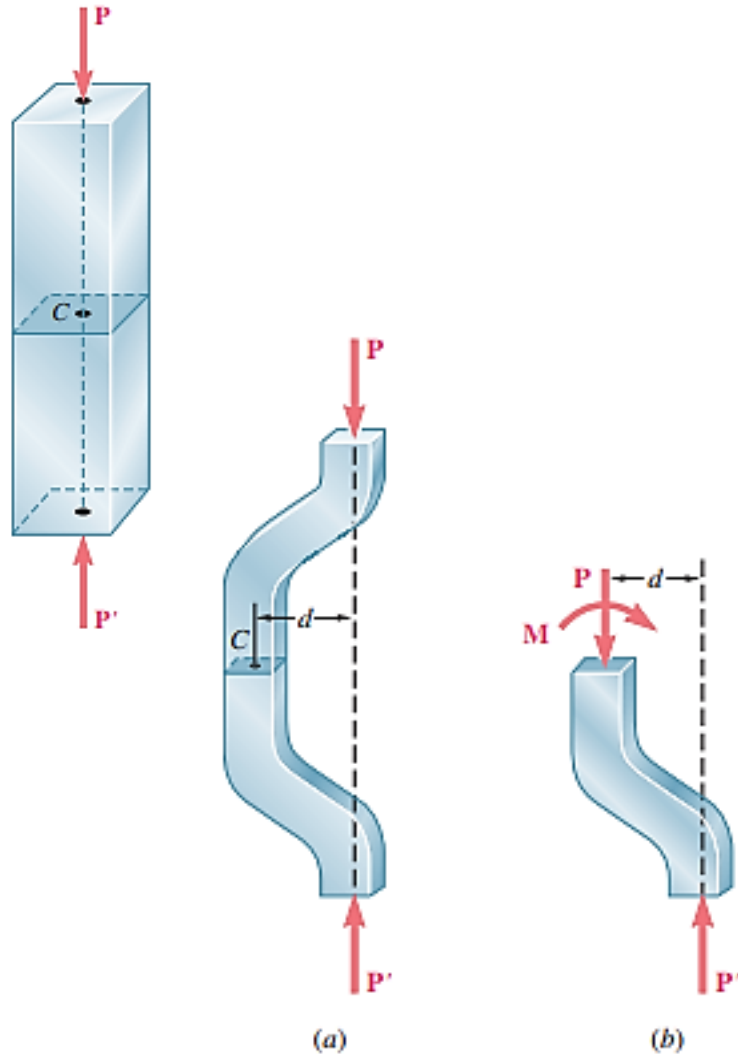


Note

- Axial force P acts through the centroid of the cross-sectional area, the stress is uniformly distributed over the cross section of the bar.



Centric & Eccentric Loading



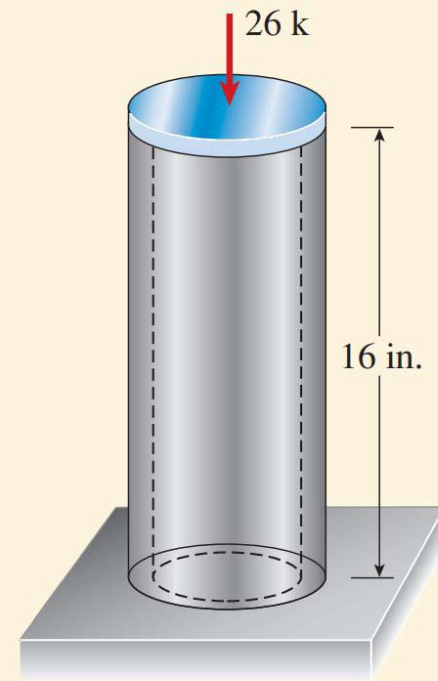
- A uniform distribution of stress in a section infers that the line of action for the resultant of the internal forces passes through the centroid of the section.
- A uniform distribution of stress is only possible if the concentrated loads on the end sections of two-force members are applied at the section centroids. This is referred to as centric loading.
- If a two-force member is eccentrically loaded, then the resultant of the stress distribution in a section must yield an axial force and a moment.
- The stress distributions in eccentrically loaded members cannot be uniform or symmetric.

Example

A short post constructed from a hollow circular tube of aluminum supports a compressive load of 26 kips (Fig. 1-5). The inner and outer diameters of the tube are $d_1 = 4.0$ in. and $d_2 = 4.5$ in., respectively, and its length is 16 in. The shortening of the post due to the load is measured as 0.012 in.

Determine the compressive stress and strain in the post. (Disregard the weight of the post itself, and assume that the post does not buckle under the load.)

Hollow
aluminum
post in
compression



Solution

Assuming that the compressive load acts at the center of the hollow tube, we can use the equation $\sigma = P/A$ (Eq. 1-1) to calculate the normal stress. The force P equals 26 k (or 26,000 lb), and the cross-sectional area A is

$$A = \frac{\pi}{4} (d_2^2 - d_1^2) = \frac{\pi}{4} [(4.5 \text{ in.})^2 - (4.0 \text{ in.})^2] = 3.338 \text{ in.}^2$$

Therefore, the compressive stress in the post is

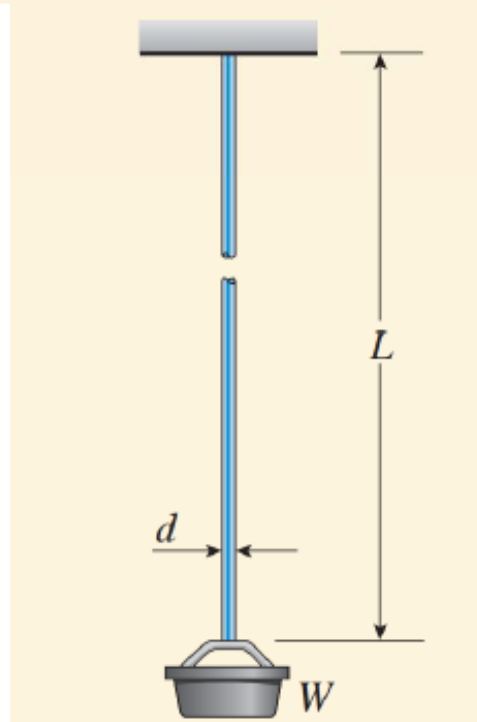
$$\sigma = \frac{P}{A} = \frac{26,000 \text{ lb}}{3.338 \text{ in.}^2} = 7790 \text{ psi}$$



Example 2

A circular steel rod of length L and diameter d hangs in a mine shaft and holds an ore bucket of weight W at its lower end (Fig. 1-6).

- (a) Obtain a formula for the maximum stress $\sigma_{\max}$ in the rod, taking into account the weight of the rod itself.
- (b) Calculate the maximum stress if $L = 40$ m, $d = 8$ mm, and $W = 1.5$ kN.



(a) The maximum axial force $F_{\max}$ in the rod occurs at the upper end and is equal to the weight W of the ore bucket plus the weight W_0 of the rod itself. The latter is equal to the weight density γ of the steel times the volume V of the rod, or

$$W_0 = \gamma V = \gamma AL \quad (1-4)$$

in which A is the cross-sectional area of the rod. Therefore, the formula for the maximum stress (from Eq. 1-1) becomes

$$\sigma_{\max} = \frac{F_{\max}}{A} = \frac{W + \gamma AL}{A} = \frac{W}{A} + \gamma L \quad (1-5) \quad \leftarrow$$

(b) To calculate the maximum stress, we substitute numerical values into the preceding equation. The cross-sectional area A equals $\pi d^2/4$, where $d = 8$ mm, and the weight density γ of steel is 77.0 kN/m^3 (from Table H-1 in Appendix H). Thus,

$$\begin{aligned}\sigma_{\max} &= \frac{1.5 \text{ kN}}{\pi(8 \text{ mm})^2/4} + (77.0 \text{ kN/m}^3)(40 \text{ m}) \\ &= 29.8 \text{ MPa} + 3.1 \text{ MPa} = 32.9 \text{ MPa}\end{aligned}$$

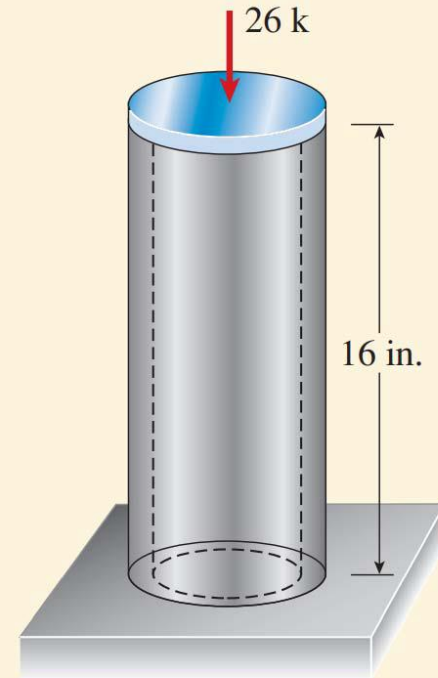

In this example, the weight of the rod contributes noticeably to the maximum stress and should not be disregarded.

Example

A short post constructed from a hollow circular tube of aluminum supports a compressive load of 26 kips (Fig. 1-5). The inner and outer diameters of the tube are $d_1 = 4.0$ in. and $d_2 = 4.5$ in., respectively,

Determine the compressive stress the post. (Disregard the weight of the post itself, and assume that the post does not buckle under the load.)

Hollow
aluminum
post in
compression



Solution

Assuming that the compressive load acts at the center of the hollow tube, we can use the equation $\sigma = P/A$ (Eq. 1-1) to calculate the normal stress. The force P equals 26 k (or 26,000 lb), and the cross-sectional area A is

$$A = \frac{\pi}{4} (d_2^2 - d_1^2) = \frac{\pi}{4} [(4.5 \text{ in.})^2 - (4.0 \text{ in.})^2] = 3.338 \text{ in.}^2$$

Therefore, the compressive stress in the post is

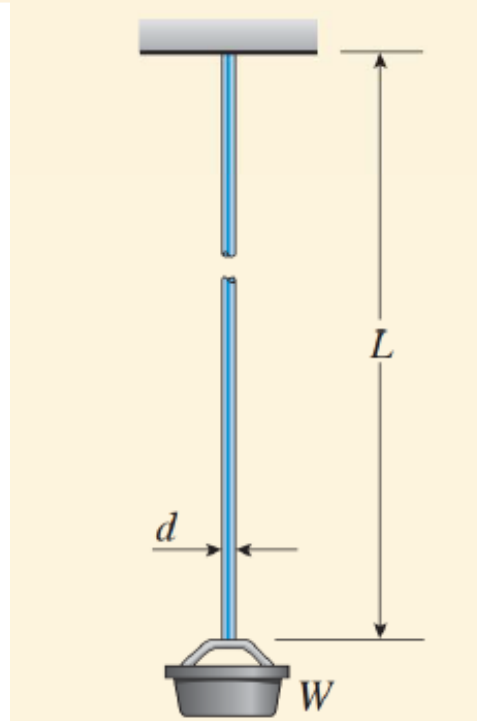
$$\sigma = \frac{P}{A} = \frac{26,000 \text{ lb}}{3.338 \text{ in.}^2} = 7790 \text{ psi}$$



Example 2

A circular steel rod of length L and diameter d hangs in a mine shaft and holds an ore bucket of weight W at its lower end (Fig. 1-6).

- (a) Obtain a formula for the maximum stress $\sigma_{\max}$ in the rod, taking into account the weight of the rod itself.
- (b) Calculate the maximum stress if $L = 40$ m, $d = 8$ mm, and $W = 1.5$ kN.



(a) The maximum axial force $F_{\max}$ in the rod occurs at the upper end and is equal to the weight W of the ore bucket plus the weight W_0 of the rod itself. The latter is equal to the weight density γ of the steel times the volume V of the rod, or

$$W_0 = \gamma V = \gamma AL \quad (1-4)$$

in which A is the cross-sectional area of the rod. Therefore, the formula for the maximum stress (from Eq. 1-1) becomes

$$\sigma_{\max} = \frac{F_{\max}}{A} = \frac{W + \gamma AL}{A} = \frac{W}{A} + \gamma L \quad (1-5) \quad \leftarrow$$

(b) To calculate the maximum stress, we substitute numerical values into the preceding equation. The cross-sectional area A equals $\pi d^2/4$, where $d = 8$ mm, and the weight density γ of steel is 77.0 kN/m^3 (from Table H-1 in Appendix H). Thus,

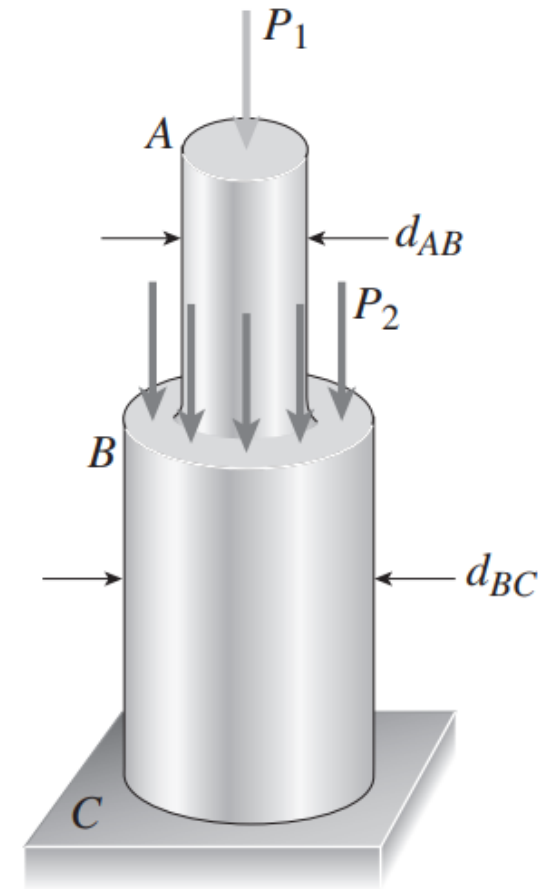
$$\begin{aligned}\sigma_{\max} &= \frac{1.5 \text{ kN}}{\pi(8 \text{ mm})^2/4} + (77.0 \text{ kN/m}^3)(40 \text{ m}) \\ &= 29.8 \text{ MPa} + 3.1 \text{ MPa} = 32.9 \text{ MPa}\end{aligned}$$


In this example, the weight of the rod contributes noticeably to the maximum stress and should not be disregarded.

Example

Problem 1.2-1 A solid circular post ABC (see figure) supports a load $P_1 = 2500$ lb acting at the top. A second load P_2 is uniformly distributed around the shelf at B . The diameters of the upper and lower parts of the post are $d_{AB} = 1.25$ in. and $d_{BC} = 2.25$ in., respectively.

- (a) Calculate the normal stress σ_{AB} in the upper part of the post.
- (b) If it is desired that the lower part of the post have the same compressive stress as the upper part, what should be the magnitude of the load P_2 ?



$$P_1 = 2500 \text{ lb}$$

$$d_{AB} = 1.25 \text{ in.}$$

$$d_{BC} = 2.25 \text{ in.}$$

(a) NORMAL STRESS IN PART AB

$$\sigma_{AB} = \frac{P_1}{A_{AB}} = \frac{2500 \text{ lb}}{\frac{\pi}{4}(1.25 \text{ in.})^2} = 2040 \text{ psi} \quad \leftarrow$$

(b) LOAD P_2 FOR EQUAL STRESSES

$$\begin{aligned} \sigma_{BC} &= \frac{P_1 + P_2}{A_{BC}} = \frac{2500 \text{ lb} + P_2}{\frac{\pi}{4}(2.25 \text{ in.})^2} \\ &= \sigma_{AB} = 2040 \text{ psi} \end{aligned}$$

$$\text{Solve for } P_2: \quad P_2 = 5600 \text{ lb} \quad \leftarrow$$

ALTERNATE SOLUTION FOR PART (b)

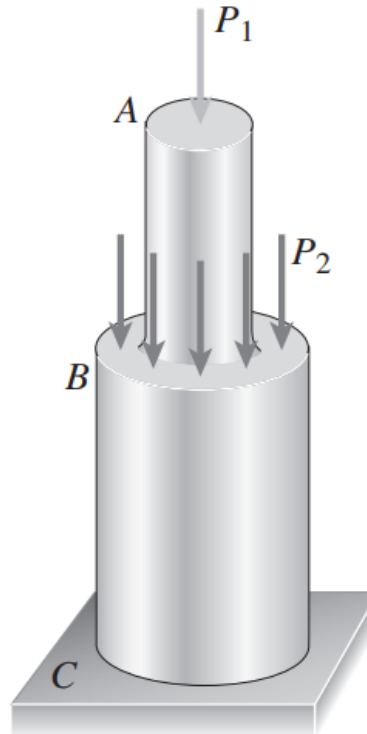
$$\sigma_{BC} = \frac{P_1 + P_2}{A_{BC}} = \frac{P_1 + P_2}{\frac{\pi}{4} d_{BC}^2}$$

$$\sigma_{AB} = \frac{P_1}{A_{AB}} = \frac{P_1}{\frac{\pi}{4} d_{AB}^2} \quad \sigma_{BC} = \sigma_{AB}$$

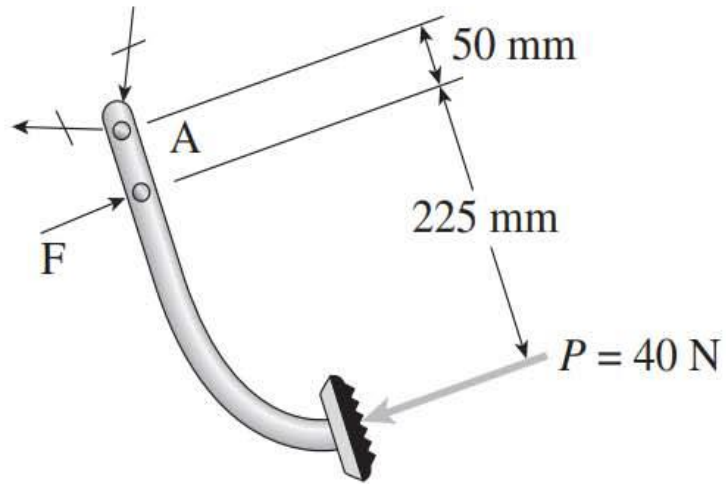
$$\frac{P_1 + P_2}{d_{BC}^2} = \frac{P_1}{d_{AB}^2} \quad \text{or} \quad P_2 = P_1 \left[\left(\frac{d_{BC}}{d_{AB}} \right)^2 - 1 \right]$$

$$\frac{d_{BC}}{d_{AB}} = 1.8$$

$$\therefore P_2 = 2.24 P_1 = 5600 \text{ lb} \quad \leftarrow$$



Solution 1.2-2 Free-body diagram of brake pedal



F = compressive force in piston rod

d = diameter of piston rod

= 5 mm

EQUILIBRIUM OF BRAKE PEDAL

$$\Sigma M_A = 0 \quad \curvearrowright \quad \curvearrowleft$$

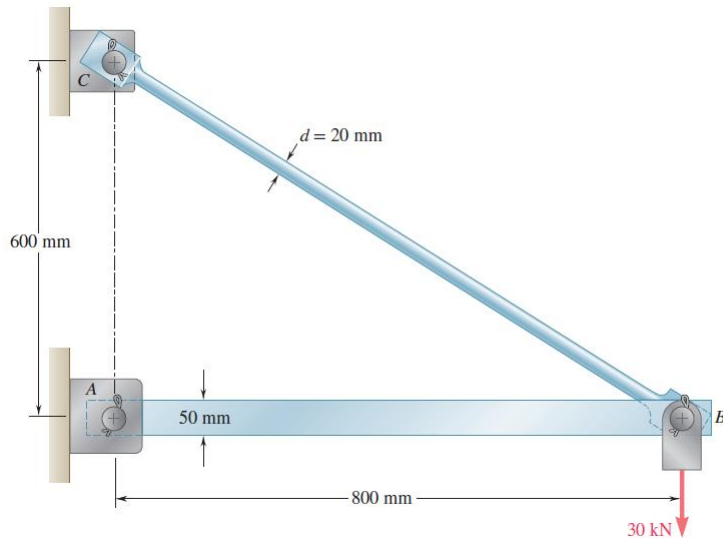
$$F(50 \text{ mm}) - P(275 \text{ mm}) = 0$$

$$F = P \left(\frac{275 \text{ mm}}{50 \text{ mm}} \right) = (40 \text{ N}) \left(\frac{275}{50} \right) = 220 \text{ N}$$

COMPRESSIVE STRESS IN PISTON ROD ($d = 5 \text{ mm}$)

$$\sigma_c = \frac{F}{A} = \frac{220 \text{ N}}{\frac{\pi}{4}(5 \text{ mm})^2} = 11.2 \text{ MPa} \quad \leftarrow$$

REVIEW OF THE METHODS OF STATICS



Consider the structure shown in Fig. 1.1, which was designed to support a 30-kN load. It consists of a boom AB with a 30×50 -mm rectangular cross section and a rod BC with a 20-mm-diameter circular cross section. These are connected by a pin at B and are supported by pins and brackets at A and C , respectively. First draw a *free-body diagram* of the structure. This is done by detaching the structure from its supports at A and C , and then showing the reactions that these supports exert on the structure (Fig. 1.2). Note that the sketch of the structure has been simplified by omitting all unnecessary details. Many of you may have recognized at this point that AB and BC are *two-force members*. For those of you who have not, we will pursue our analysis, ignoring that fact and assuming that the directions of the reactions at A and C are unknown. Each of these reactions are represented by two components: A_x and A_y at A , and C_x and C_y at C . The three equilibrium equations are.

$$+\circlearrowleft \Sigma M_C = 0: \quad A_x(0.6 \text{ m}) - (30 \text{ kN})(0.8 \text{ m}) = 0$$

$$A_x = +40 \text{ kN} \quad (1.1)$$

$$+\rightarrow \Sigma F_x = 0: \quad A_x + C_x = 0$$

$$C_x = -A_x \quad C_x = -40 \text{ kN} \quad (1.2)$$

$$+\uparrow \Sigma F_y = 0: \quad A_y + C_y - 30 \text{ kN} = 0$$

$$A_y + C_y = +30 \text{ kN} \quad (1.3)$$

REVIEW OF THE METHODS OF STATICS

We have found two of the four unknowns, but cannot determine the other two from these equations, and no additional independent equation can be obtained from the free-body diagram of the structure. We must now dismember the structure. Considering the free-body diagram of the boom AB (Fig. 1.3), we write the following equilibrium equation:

$$+\circlearrowleft \Sigma M_B = 0: \quad -A_y(0.8 \text{ m}) = 0 \quad A_y = 0 \quad (1.4)$$

Substituting for A_y from Eq. (1.4) into Eq. (1.3), we obtain $C_y = +30 \text{ kN}$. Expressing the results obtained for the reactions at A and C in vector form, we have

$$A = 40 \text{ kN} \rightarrow \quad C_x = 40 \text{ kN} \leftarrow \quad C_y = 30 \text{ kN} \uparrow$$

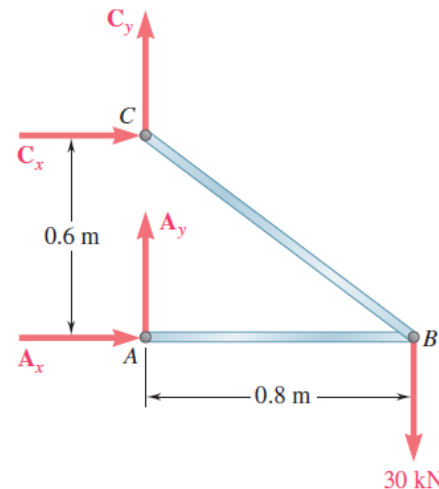


Fig. 1.2 Free-body diagram of boom showing applied load and reaction forces.

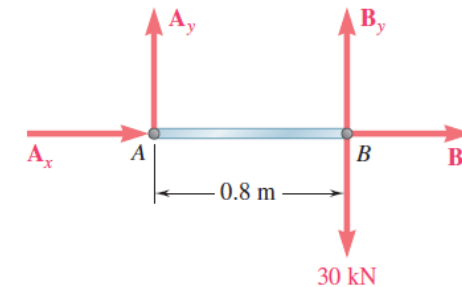


Fig. 1.3 Free-body diagram of member AB freed from structure.

REVIEW OF THE METHODS OF STATICS

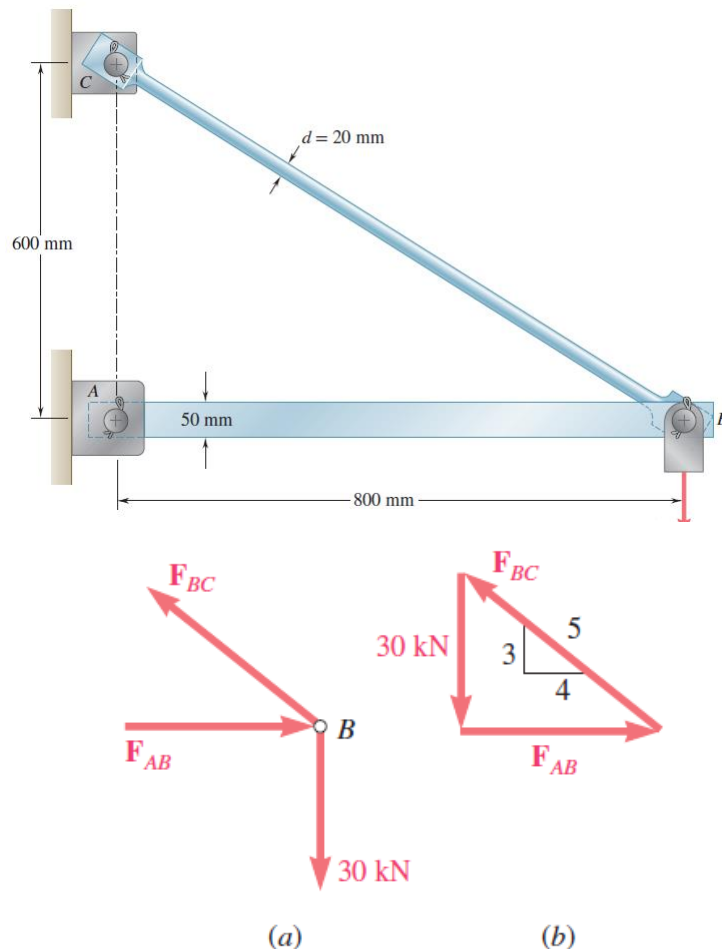


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

Note that the reaction at A is directed along the axis of the boom AB and causes compression in that member. Observe that the components C_x and C_y of the reaction at C are, respectively, proportional to the horizontal and vertical components of the distance from B to C and that the reaction at C is equal to 50 kN, is directed along the axis of the rod BC , and causes tension in that member.

These results could have been anticipated by recognizing that AB and BC are two-force members, i.e., members that are subjected to forces at only two points, these points being A and B for member AB , and B and C for member BC . Indeed, for a two-force member the lines of action of the resultants of the forces acting at each of the two points are equal and opposite and pass through both points. Using this property, we could have obtained a simpler solution by considering the free-body diagram of pin B . The forces on pin B , F_{AB} and F_{BC} , are exerted, respectively, by members AB and BC and the 30-kN load (Fig. 1.4a). Pin B is shown to be in equilibrium by drawing the corresponding force triangle (Fig. 1.4b).

Since force F_{BC} is directed along member BC , its slope is the same as that of BC , namely, $3/4$. We can, therefore, write the proportion

REVIEW OF THE METHODS OF STATICS

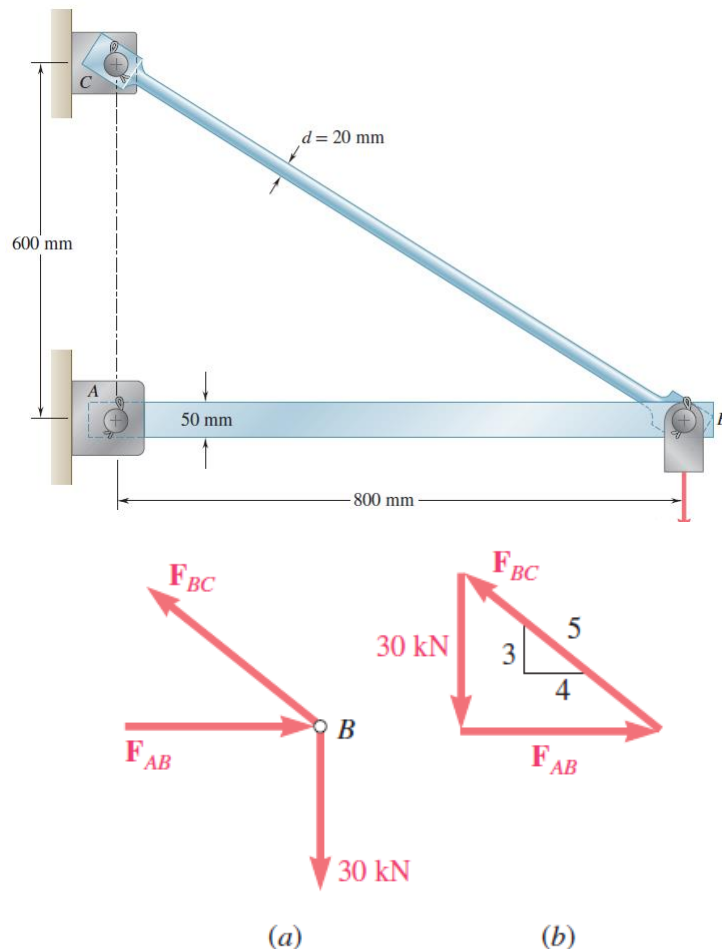


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

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These results could have been anticipated by recognizing that AB and BC are two-force members, i.e., members that are subjected to forces at only two points, these points being A and B for member AB , and B and C for member BC . Indeed, for a two-force member the lines of action of the resultants of the forces acting at each of the two points are equal and opposite and pass through both points. Using this property, we could have obtained a simpler solution by considering the free-body diagram of pin B . The forces on pin B , F_{AB} and F_{BC} , are exerted, respectively, by members AB and BC and the 30-kN load (Fig. 1.4a). Pin B is shown to be in equilibrium by drawing the corresponding force triangle (Fig. 1.4b).

Since force F_{BC} is directed along member BC , its slope is the same as that of BC , namely, $3/4$. We can, therefore, write the proportion

REVIEW OF THE METHODS OF STATICS

$$\frac{F_{AB}}{4} = \frac{F_{BC}}{5} = \frac{30 \text{ kN}}{3}$$

from which

$$F_{AB} = 40 \text{ kN} \quad F_{BC} = 50 \text{ kN}$$

Forces $\mathbf{F}'_{AB}$ and $\mathbf{F}'_{BC}$ exerted by pin B on boom AB and rod BC are equal and opposite to $\mathbf{F}_{AB}$ and $\mathbf{F}_{BC}$ (Fig. 1.5).

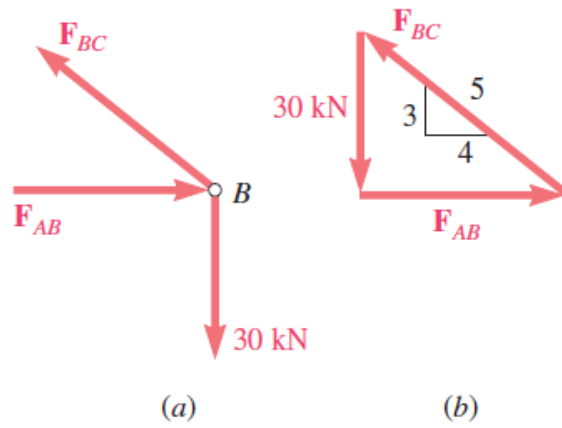


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

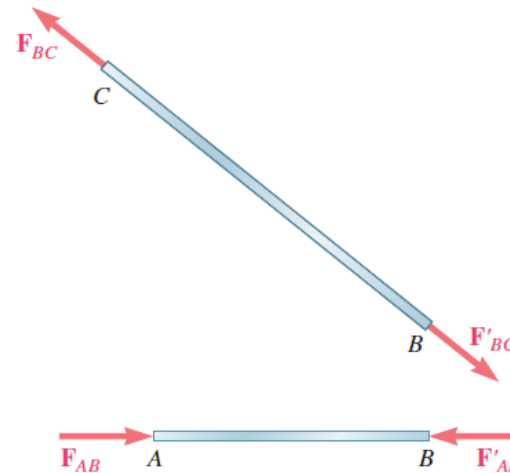


Fig. 1.5 Free-body diagrams of two-force members AB and BC .

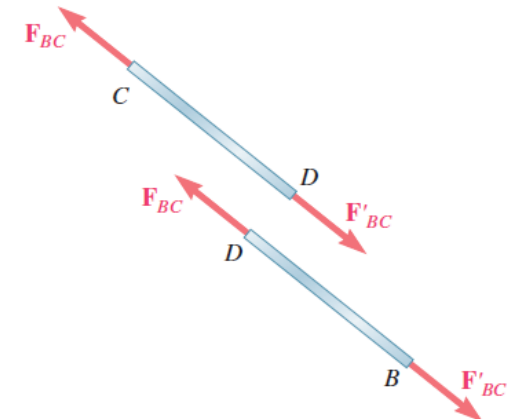
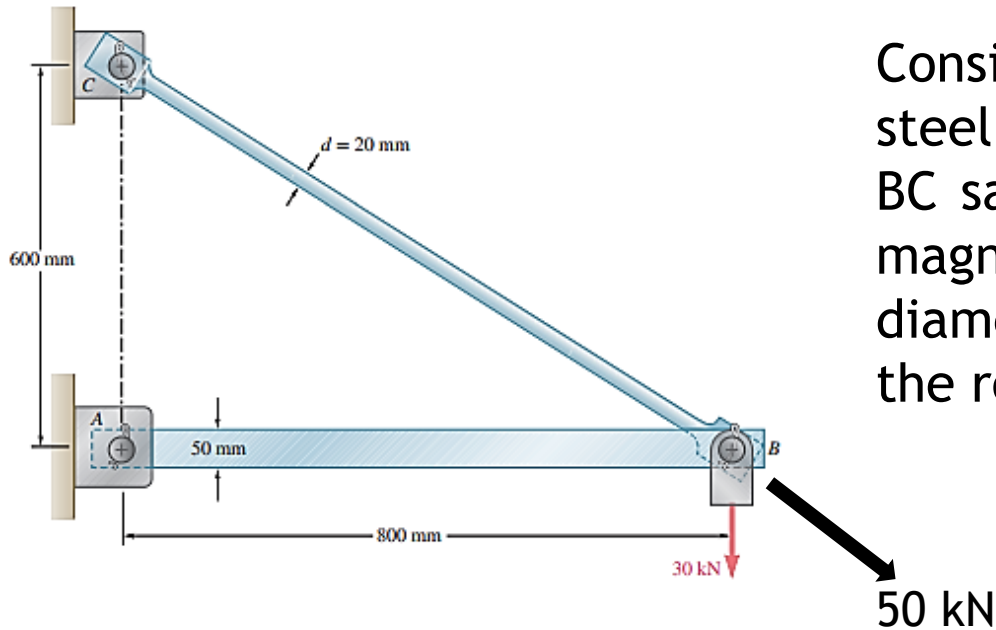


Fig. 1.6 Free-body diagrams of sections of rod BC .

Example:1



Considering the structure of , assume that rod BC is made of a steel with a maximum allowable stress $\sigma_{all} = 165 \text{ MPa}$. Can rod BC safely support the load to which it will be subjected? The magnitude of the force F_{BC} in the rod was 50 kN. If the diameter of the rod is 20 mm. Determine the stress created in the rod by the given loading.

$$P = F_{BC} = +50 \text{ kN} = +50 \times 10^3 \text{ N}$$

$$A = \pi r^2 = \pi \left(\frac{20 \text{ mm}}{2} \right)^2 = \pi (10 \times 10^{-3} \text{ m})^2 = 314 \times 10^{-6} \text{ m}^2$$

$$\sigma = \frac{P}{A} = \frac{+50 \times 10^3 \text{ N}}{314 \times 10^{-6} \text{ m}^2} = +159 \times 10^6 \text{ Pa} = +159 \text{ MPa}$$

Since σ is smaller than σ_{all} of the allowable stress in the steel used, rod BC can safely support the load.



Thank you

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